TinyAudit: On-Device, Uncertainty-Aware Fairness Audits that Survive (and Expose) Compression

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Abstract

Machine learning is moving onto microcontroller-class hardware, where models are aggressively compressed (pruned, quantized) to fit. Responsible-AI audits, in contrast, are almost always run on full-precision models on a workstation, and they typically look at fairness, uncertainty, and explainability in isolation. We present TinyAudit, an auditing pipeline that measures all three together on the compressed artifact rather than the original, and that profiles its own cost. Theory already tells us calibration and parity cannot generally agree, and that compression harms underrepresented groups; we claim neither as new. Our contribution is measuring what those results do to a deployed model, reported as means over 10 seeds. First, the two lenses do not merely disagree, they select different groups: the ordering by selection rate and the ordering by per-group calibration error cross on UCI Adult and again on COMPAS, whose favorable direction is inverted. Second, and more actionable, compression moves the two in opposite directions: pruning an MLP to 90% sparsity drives its demographic-parity difference down roughly 12-fold while per-group calibration error climbs roughly 10-fold, so a parity-only audit run after compression ranks the most degraded model in the sweep as the fairest. Third, the audit's own peak working set is about 0.6 MB plus 2.3 KB per sample, constant in test-set size when streamed. That constant does not yet fit the 32 to 512 KB parts at the low end of the TinyML range, and we report the gap rather than claim it closed.

1 Introduction

TinyML runs inference on microcontrollers with 32 to 512 KB of SRAM [19]. To fit that budget, deployed models are pruned and quantized. The auditing tools that would check those models for fairness were built for a different world: cloud notebooks, full-precision weights, and demographic labels available at test time [14, 2, 8]. Existing TinyML benchmarks measure latency, accuracy, and energy, but not fairness or explainability [19]. The result is a gap. The models most likely to be compressed are the least likely to be audited, and when they are audited, the audit sees a model that is not the one deployed.

A second problem is that fairness audits usually stop at point predictions. A model can look fair on selection rates while its uncertainty is biased against a group [13], and abstaining on high-uncertainty inputs does not by itself repair group imbalance [17]. Uncertainty quantification now fits on KB-sized devices [7], but no existing system evaluates those estimates through a fairness lens, and no fairness toolkit fits on the device.

TinyAudit closes both gaps at once: a seven-stage pipeline that measures fairness, uncertainty, and explainability on the compressed model, inside a memory envelope sized for the device being audited.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Relation to known results. Both results have theoretical ancestors and we want to be precise about what is new. That calibration and parity-style criteria cannot generally hold at once is established [11, 16], in Chouldechova's case on COMPAS specifically [4]; that compression disproportionately damages underrepresented groups is likewise documented [9, 3]. We claim neither as new. What is new is the setting: those results come from full-precision models and workstation tooling, and the model deployed to a microcontroller is neither. The two interact in a way that matters operationally. Compression moves parity and calibration in *opposite* directions, so the metric most audits report improves as the model gets worse, and no toolkit measures this on the compressed artifact inside the device's own budget.

Contributions:

1. An on-device audit instrument: a seven-stage pipeline running fairness, uncertainty, and explainability on the compressed model, emitting a one-page traffic-light card and a machine-readable manifest.
2. A measurement showing the calibration/parity tension is group-selective in practice on the deployed model: the two lenses rank sensitive groups differently on Adult and replicate on COMPAS, whose favorable direction is inverted (Section 4).
3. A concrete failure mode for compression-era audits: under pruning, demographic parity falls roughly 12-fold while per-group calibration error rises roughly 10-fold, so a parity-only audit ranks the most degraded model as the fairest (Section 5).
4. A feasibility measurement of the audit itself, which to our knowledge has not been reported before: peak working set is about 0.6 MB plus 2.3 KB per sample and is constant in test-set size when streamed (Section 6).

2 The TinyAudit instrument

TinyAudit takes a fitted model and a labeled test set and produces a one-page audit card plus a JSON manifest that every stage writes to. The card is generated from the manifest and never hand-edited; every figure in it is backed by a committed CSV. The stages, in order:

1. Profile. Parameter count, an analytic FLOPs proxy, peak RAM (via `tracemalloc`), and wall-clock per sample. **2. Point fairness.** Demographic-parity difference (DP), equalized-odds difference (EO), and disparate-impact ratio (DI), each with an explicitly pinned favorable direction and a hypothesis-tested range. **3. Uncertainty.** MC dropout (MLP only), a lightweight perturbation ensemble, and a QUTE-style early-exit estimator [7]; each returns predictive entropy, variance, and mutual information. **4. Uncertainty-aware fairness.** Per-group predictive entropy, per-group expected calibration error (ECE), and a selective-fairness AUC that integrates parity as low-confidence predictions are progressively abstained. **5. Explainability.** A dependency-free occlusion explainer reporting per-group feature importance and which features enter one group's top-k but not another's. SHAP and LIME wrappers ship with the package but stay out of the audit path, being costly and fragile at the edge [18, 15]. The stage is counted in the footprint of Section 6; it yields no fairness result here (Section 7). **6. Compression.** int8 dynamic quantization or magnitude pruning, applied *before* every metric above, so the audit sees the deployed model. **7. Render.** The one-page, traffic-light-coded card.

The audit's working set is dominated by per-sample buffers, so it streams in small batches at a peak RAM independent of test-set size (Section 6). The whole public API is one call:

```
from tinyaudit import audit
card = audit(model=clf, data=(X_test, y_test), sensitive="sex",
             methods=["fairness","uncertainty","xai"], compression="int8")
card.to_pdf("adult_lr_int8.pdf")
```

3 Experimental setup

Datasets. UCI Adult [12] (36,631/12,211 train/test, 101 features), COMPAS [1] (4,629/1,543, 5 features; here the positive outcome is being flagged high-risk, so higher is worse), and Folktables ACSIncome [5] (146,748/48,917, 8 features, 2018 California). Sensitive attributes are sex and race

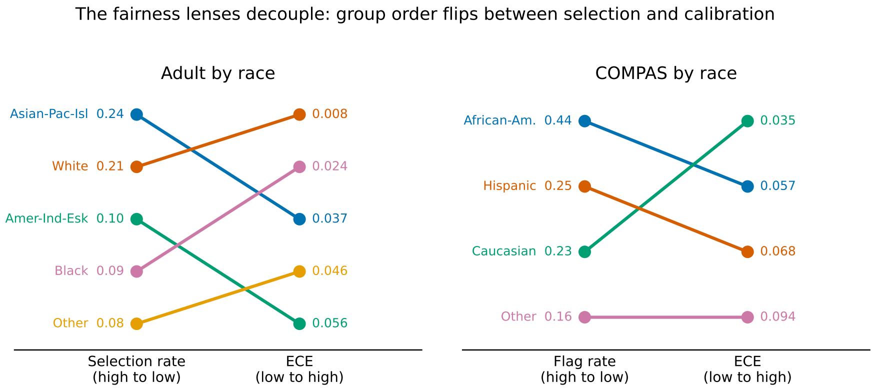

Figure 1: Finding 1. Rank slope charts for Adult (left) and COMPAS (right) by race, uncompressed logistic model, means over 10 seeds. Each line joins a group's rank by selection/flag rate to its rank by per-group ECE. Crossing lines mean the two lenses order the groups differently.

throughout. The first two are the near-universal fairness benchmarks [6]; ACSIncome is the modern replacement for Adult's 1994 distribution [5].

Models. Scikit-learn estimators on standardized features: logistic regression, an MLP with two hidden layers of 32 units, and a decision tree. Scaling matters, since without it the Adult MLP sits near the majority-class rate (0.54 accuracy) versus 0.84 with it. The scaler is fit on train only and applied outside the model, so compression and perturbation reach the raw weights.

Reproducibility. A single seed flag seeds `random`, NumPy, torch, and `PYTHONHASHSEED`. Every headline number is a mean $\pm$ std over seeds 0 to 9. Determinism is guaranteed on Linux/x86; elsewhere we expect last-digit float drift, which the std columns absorb. All models land in published ranges (Adult logistic regression 0.852 ± 0.003 accuracy at ROC-AUC 0.906 ± 0.002, Adult MLP 0.837 ± 0.003, COMPAS logistic 0.680 ± 0.006, ACSIncome MLP 0.808 ± 0.003; the full grid is a committed CSV). The fairness results therefore sit on models that work, not on degenerate predictors.

4 Result 1: the two lenses select different groups

Point-prediction fairness asks how often each group receives the favorable outcome. Calibration fairness asks how honest the model's confidence is per group. The impossibility results [11, 4] say these cannot generally agree; what they do not say is *which* group each lens singles out on a given deployed model, and that is what an auditor actually has to act on. On the uncompressed logistic model, the two lenses rank the sensitive groups differently, and the disagreement is not a uniform offset but a reordering (Figure 1).

Adult by race (five groups). Selection rate orders them Asian-Pacific-Islander (0.240 ± 0.014), White (0.207 ± 0.005), American-Indian-Eskimo (0.098 ± 0.026), Black (0.091 ± 0.008), Other (0.081 ± 0.024). Calibration orders them differently: White is only second-most-selected yet by far the best calibrated (ECE 0.008 ± 0.002, three to seven times lower than every other group), while American-Indian-Eskimo is worst (0.056 ± 0.019). The smaller groups carry higher, noisier calibration error that the selection view never surfaces. **By sex** the disagreement inverts: a large selection gap (Male 0.253 ± 0.007 vs Female 0.079 ± 0.004) with negligible calibration difference (ECE 0.009 ± 0.003 vs 0.014 ± 0.004).

COMPAS by race (positive outcome = flagged high-risk, so higher is worse; groups with $n < 10$ dropped). African-American defendants are flagged at roughly twice the rate of the next group (0.441 ± 0.015 vs Hispanic 0.249 ± 0.046), the well-known point disparity. Yet the worst-calibrated group, Other (ECE 0.094 ± 0.029), is flagged *least* (0.164 ± 0.033), and the best calibrated is Caucasian (0.035 ± 0.016). By sex, males are flagged more (0.362 ± 0.008 vs 0.223 ± 0.018) while females are markedly worse calibrated (0.085 ± 0.016 vs 0.033 ± 0.005). The reordering therefore holds on a second dataset whose favorable direction is inverted, so it is not an Adult-specific label-convention artifact.

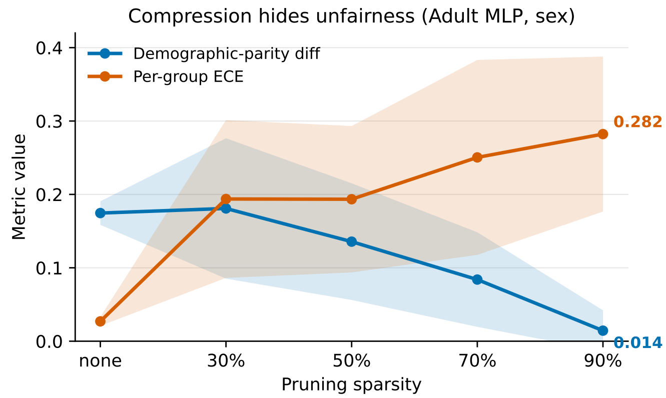

Figure 2: Finding 2. Adult MLP by sex under magnitude pruning. Demographic-parity difference falls while per-group ECE climbs; bands are ± 1 std over 10 seeds. The bands overlap at intermediate sparsities, so the robust claim is the endpoint contrast, not the middle ordering.

Why it matters on-device. Sensitive labels are rarely available at inference time, so on-device audit triggers lean on confidence scores as a proxy [10]. If the lenses disagree about which group is worst served, a uniform confidence threshold can track the wrong disparity, consistent with uniform abstention amplifying imbalance [17] and point metrics masking biased uncertainty [13]. Reporting both is the cheap fix, and TinyAudit does so by construction.

5 Result 2: compression can hide unfairness

We prune the Adult MLP to increasing sparsity and re-run the full audit at each level (Figure 2). By sex, the demographic-parity difference falls from 0.175 ± 0.016 (unpruned) to 0.014 ± 0.028 at 90% sparsity, a roughly 12-fold drop, while per-group ECE climbs from 0.027 ± 0.006 to 0.282 ± 0.106, a roughly 10-fold rise. By race the shape is the same: DP $0.150 \pm 0.024 \rightarrow 0.019 \pm 0.038$, ECE $0.051 \pm 0.007 \rightarrow 0.314 \pm 0.108$. The model looks fairer on parity precisely as its calibration collapses.

The mechanism is degradation toward a near-constant predictor, and we should be clear that the endpoint of that mechanism is textbook. A constant classifier trivially satisfies demographic parity, which is the standard counterexample for why parity alone is insufficient [16]. Our point is not that this endpoint exists but that a routine compression pipeline walks a model toward it gradually and silently, and that the audit metric most commonly reported improves monotonically the whole way. Nothing in a parity-only report distinguishes a model that became fairer from one that stopped discriminating because it stopped predicting. The corner case makes the endpoint vivid: the COMPAS logistic model at 90% pruning (only about 5 features to begin with) predicts a single class for everyone and reports DP $= 0.000$, DI $= 1.000$, perfect parity by construction. We read those cells as an illustration of the mechanism, not a trend.

Two qualifications. Intermediate pruning levels are seed-noisy (parity std 0.06 to 0.10), so the robust claim is the direction between endpoints. And the effect is strongest where pruning actually damages the model: the logistic model barely degrades (DP $0.174 \pm 0.008 \rightarrow 0.126 \pm 0.014$ at 90%) and shows only a mild version of it.

The upshot for auditors: an audit that checks only demographic parity, run only after compression, would report the pruned model as the *fairest* in the sweep. An uncertainty-aware audit flags the same model as the most broken.

6 Feasibility: what the audit itself costs

Audit tooling is normally not profiled at all: the model's footprint is reported and the auditor's is assumed free because it runs on a workstation. If the audit is to run where the model runs, that

Table 1: Audit pipeline footprint (the pipeline itself, not the audited model), profiled with `tracemalloc` over 3 seeds. Peak RAM is governed by the streaming batch size, not the test-set size.

Dataset	Model	Batch	Audit peak RAM (KB)	Audit wall-clock (s)
Adult	logreg	64	610	0.03
Adult	logreg	256	641	0.06
Adult	logreg	1024	2,477	0.11
Adult	logreg	4096	9,826	1.10
Adult	MLP	256	946	0.08
Adult	MLP	4096	9,824	1.28
COMPAS	logreg	256	609	0.01
COMPAS	MLP	1543	1,181	0.05

assumption has to be checked. We profile the full three-method audit with `tracemalloc` and report the pipeline's own peak RAM and wall-clock, not the audited model's (Table 1).

The working set is dominated by per-sample buffers: the feature matrix, the ensemble's stacked prediction arrays, and the explainer's perturbation batch. Fitting the Adult endpoints (625,005 bytes at batch 64 to 10,061,660 bytes at batch 4096) gives

$$\text{peak RAM} \approx 0.6\,\text{MB} + 2.3\,\text{KB} \times \text{batch size}. \tag{1}$$

The structural claim is the one we would defend: an arbitrarily large test set streams through the audit in fixed-size batches at constant peak RAM, so audit cost is set by the batch size the engineer picks, not by how much test data exists.

The absolute constant needs more care. At a 256-row batch the audit peaks at 0.61 to 0.95 MB in under 0.1 s. That does *not* fit the 32 to 512 KB parts at the low end of the TinyML range, and we do not claim it does; it lands within reach of the high end, such as Cortex-M7 parts with roughly 1 MB of SRAM. Equation 1 shows why smaller batches help only so much: the engineer controls the per-sample term, not the 0.6 MB fixed cost, and driving that fixed cost below 512 KB is the concrete target this measurement identifies. Two caveats bound the number further. The audited Adult MLP itself peaks at 6.1 MB, so here it stands in for an MCU-class model rather than being one, with the 62 KB COMPAS logistic model the realistic case; and `tracemalloc` excludes the interpreter while counting Python object overhead a compiled port would not pay, so this is neither a clean upper nor lower bound on that port. What survives is the scaling law, not the constant.

7 Limitations

Features are standardized before fitting, so the numbers describe the scaled-input regime. The uncertainty ensemble is five lightly-jittered copies of the compressed model, not five retrains, so uncertainty magnitudes are relative. int8 models' float weights are unreachable, so their uncertainty columns are blank; decision trees have no dense weights and sit out the compression grid. Footprint numbers are measured on x86, not on a microcontroller. The explainability stage yields no fairness result. On uncompressed Adult the model does put different features in different groups' top-k, but under pruning that divergence is non-monotone and within seed noise (MLP by race, 22.6 ± 2.9 flips to 18.3 ± 5.5 at 90%), it does not replicate on COMPAS, and the pruned logistic model's apparent collapse to zero flips is an artifact of 90% sparsity leaving as many nonzero weights as the shortlist has slots. We report that stage as instrument and cost, not evidence. Neither result we do report is theoretically novel [11, 4, 9]; the contribution is measuring them on the model that ships.

8 Conclusion

Fairness toolkits assume the cloud [14, 2, 8] and TinyML benchmarks ignore fairness [19]. The theory predicting our results predates us [11, 4, 9]; what we add is that those effects are measurable, group-selective, and actively misleading on the model that actually ships, at under a megabyte per streaming batch. The warning is narrow and hard to dismiss: an audit reporting parity alone, run after compression, ranks the most degraded model in a sweep as the fairest. Closing the gap to 512 KB and porting to hardware is next work.

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